STATISTICAL ARBITRAGE & BACKTESTING

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1. Introduction

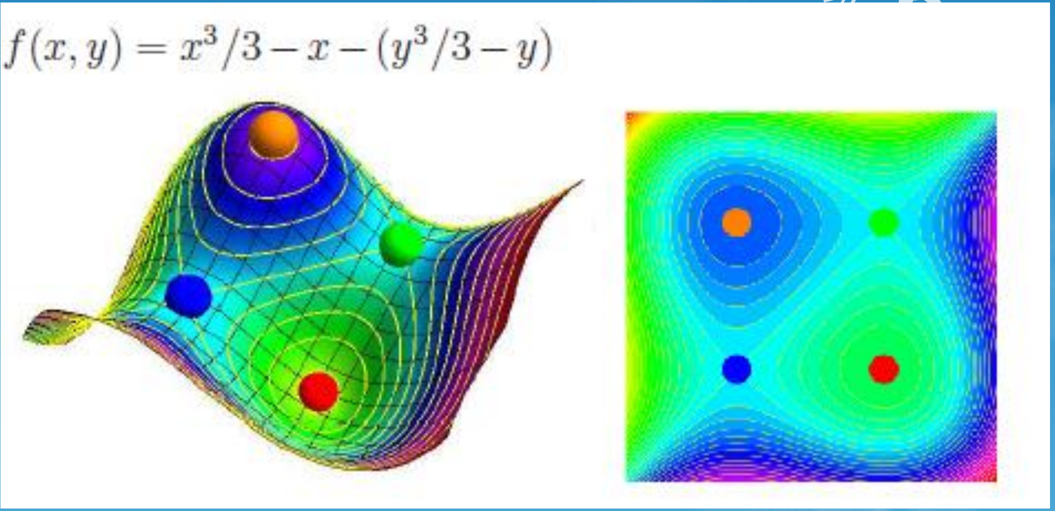
What is a “Quant”

- “Quant” is the short version of “Quantitative Analyst”
- They are the “math gurus” in a financial firm e.g. bank or fund
  - They build the mathematical tools for:
    - Pricing
    - Underwriting
    - Trading
    - Risk management
    - Etc
- Three types of quants:
  - “Probability” quants
  - “Statistics” quants
  - “Machine Learning” quants

harringtonstarr.com

"Probability" Quant

- They provide the **mathematical models** for the firm
  - Commonly referred as "modellers"
- They need a strong background in Mathematics:
  - Probability
  - Differential equations
    - ODEs
    - PDEs
    - SDEs
  - Numerical methods
  - Optimisation
- And some Economics:
  - Efficient Market Theory
  - Utility functions
  - Cost functions
  - Mean-variance optimisation



math.harvard.edu

"Statistics" Quant

- They provide the **data & analytic tools** for the firm
  - Commonly referred as "number crunchers"
- They need a strong background in Statistics:
  - Data mining
  - Data crunching
  - Distributions
  - Curve fitting
  - Tests of hypotheses
- And some Economics:
  - Econometrics
  - Macro- and Microeconomics
  - Price-formation processes

seleritysas.com

“Machine Learning” Quant

- They build **large datasets and models to predict behaviours and choices**
- ML started as a subset of Statistics with a Computer Science flavour
  - But it has already become a discipline on its own
- The Machine Learning models deal with:
  - Large datasets and large number of parameters
  - Recalibration of parameters based on the outcome of previous decisions
- Different ML models to predict and forecast:
  - Linear regression
  - Logistic regression
  - Neural Networks
- Deep Learning is used almost everywhere
  - It is a scientific field on its own
  - Separated from ML and AI

hotforexblog.com

2. Statistical Arbitrage

Statistical Arbitrage

- **Principle 1**
  - There are identifiable patterns in the financial markets
- That means:
  - We can find exploitable trading strategies
- Example:
  - Price trends
  - Volatility trends
  - Volume trends
  - Macro events
  - News

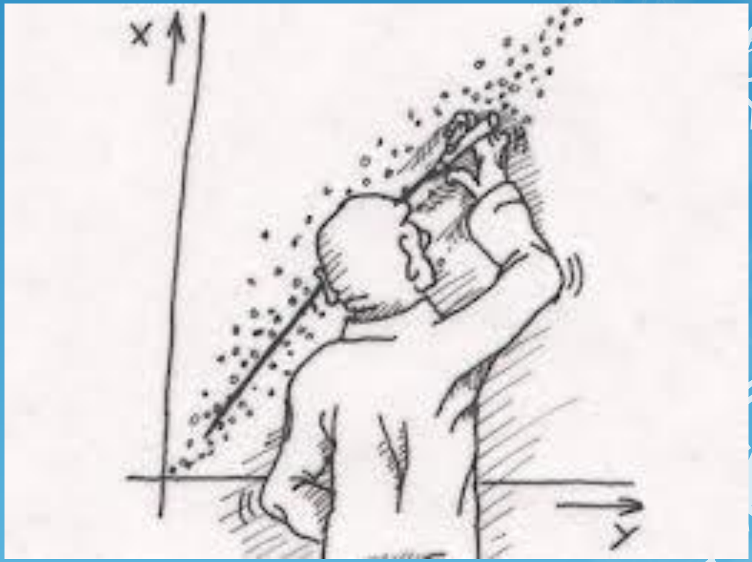
acting-man.com

Statistical Arbitrage

- **Principle 2**
  - Some identified patterns are statistically robust

- That means:
  - Some patterns are stable under small changes on their input and their parameters

- Example:
  - Parametric models:
    - Model is stable
  - Non-parametric models:
    - Distribution is stable
  - Recurrent behaviour:
    - Volumes and volatility spikes



algo-trades.com

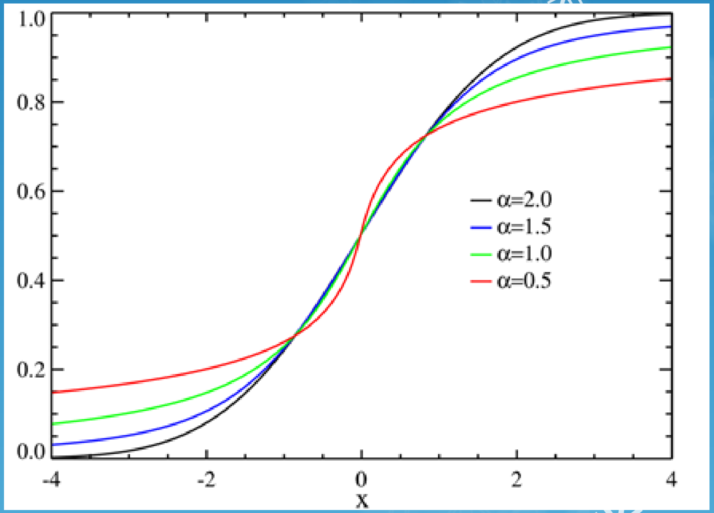
Statistical Arbitrage

- **Principle 3**
  - For some patterns, past behaviour can on average predict future behaviour

- That means:
  - Some patterns are stable in time
  - At least in the short term
  - Potential need of “periodic recalibration”

- Example:
  - Volume and volatility curves
  - Correlations

Stability of cumulative distribution



wikipedia.com

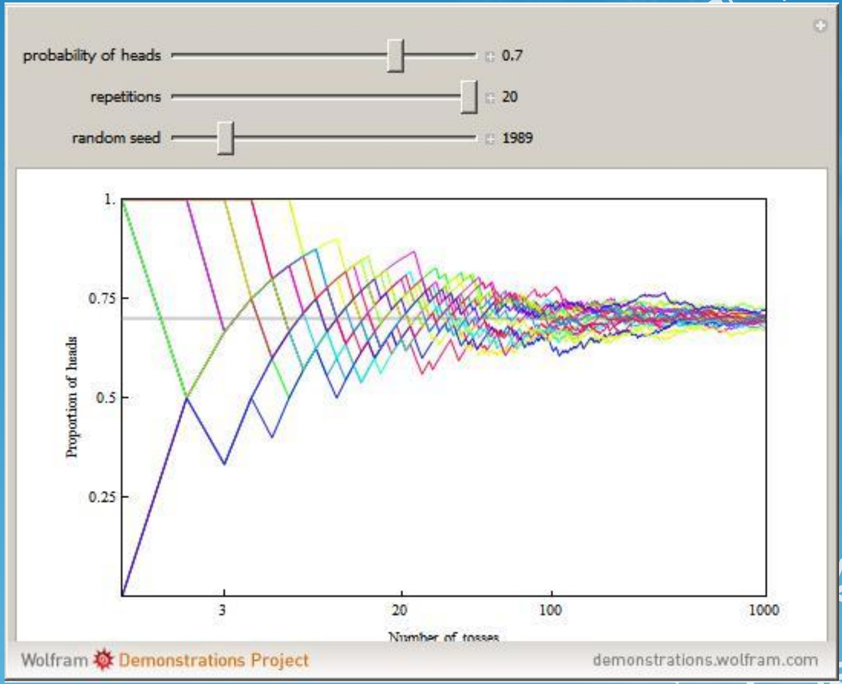
- Predicting future prices can be possible but is very subtle

Statistical Arbitrage

- **Principle 4**
  - A strategy exploiting a robust pattern is profitable <u>on average</u>

- That means:
  - Do not expect to win all the time
  - But you can win in the long run:
    - Law of Large Numbers
    - Central Limit Theorem

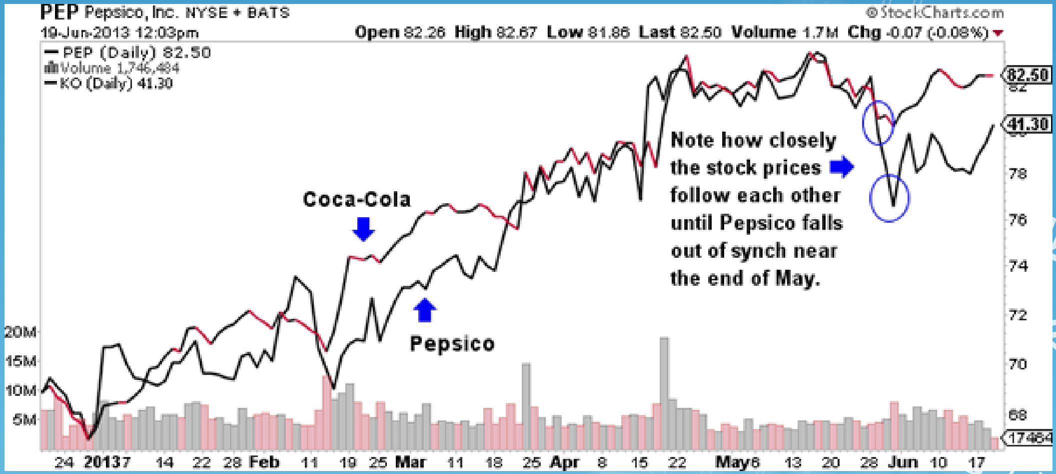
- Example:
  - Lottery tickets
  - Insurance premium
  - Option pricing
  - Market-making



wolfram.com

Statistical Arbitrage

- **Principle 5**
  - A pattern normally changes after some time
- That means:
  - Even if patterns are stable in time, they do not to last forever
  - Some patterns can disappear if the market changes
  - Frequent recalibration to determine when a strategy is no longer profitable
- Example:
  - Correlation strategies:
    - Pair trading
    - Index arb



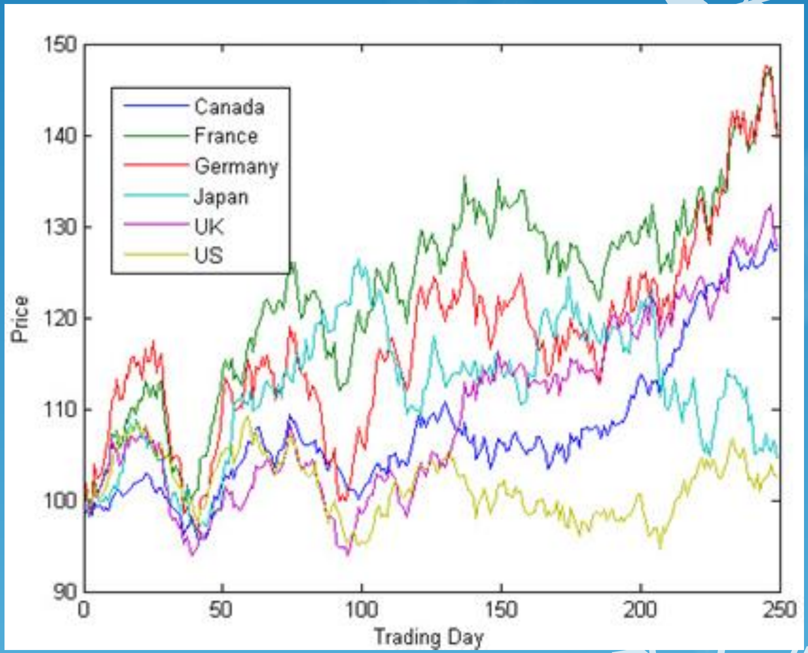
nasdaq.com

3. Backtesting

Backtesting

- Stage 1
  - Build prototype of the trading strategy

- What to do:
  - Code the rules of the algorithm
    - Matlab, R or Python
  - Simulate time series:
    - Monte Carlo
  - Use simulations to test:
    - Code
    - Rules
    - Dependence to parameters
  - Get a first glimpse of the distribution

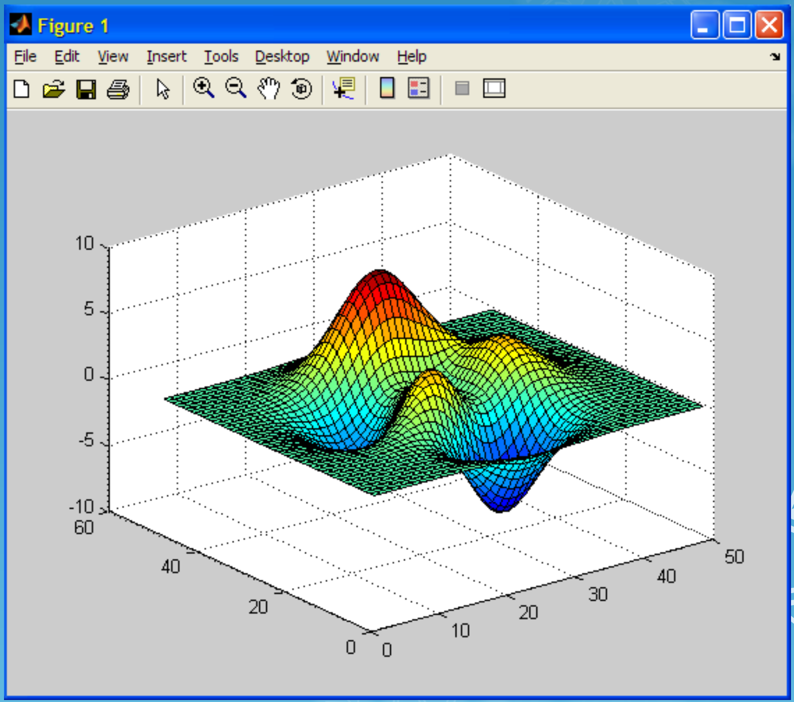


mathworks.com

Backtesting

- Stage 2
  - Define the parameters and the "utility function"

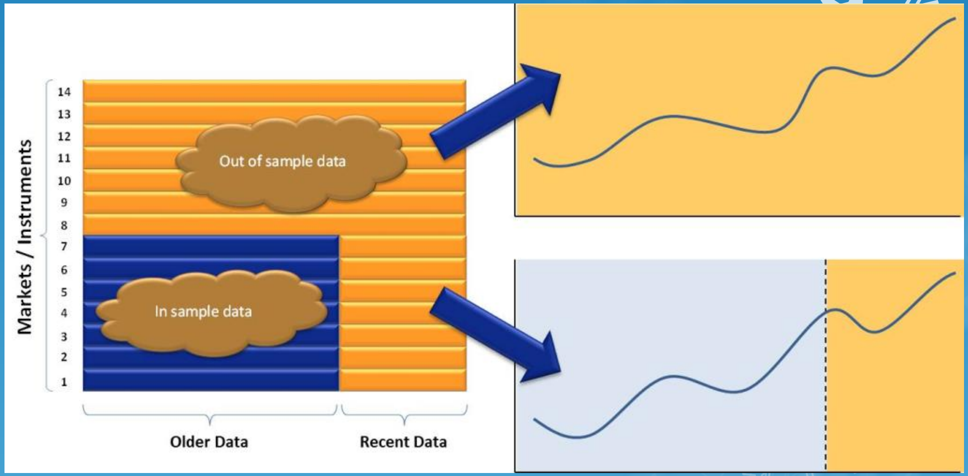
- What to do:
  - Define the space of parameters
    - Potentially reduce dimensions
  - Define the optimisation function
    - Utility function
      - Maximise
    - Cost function
      - Minimise



Matlab

Backtesting

- Stage 3
  - Define your “In Sample” and “Out of Sample”



tradingsystemlife.com

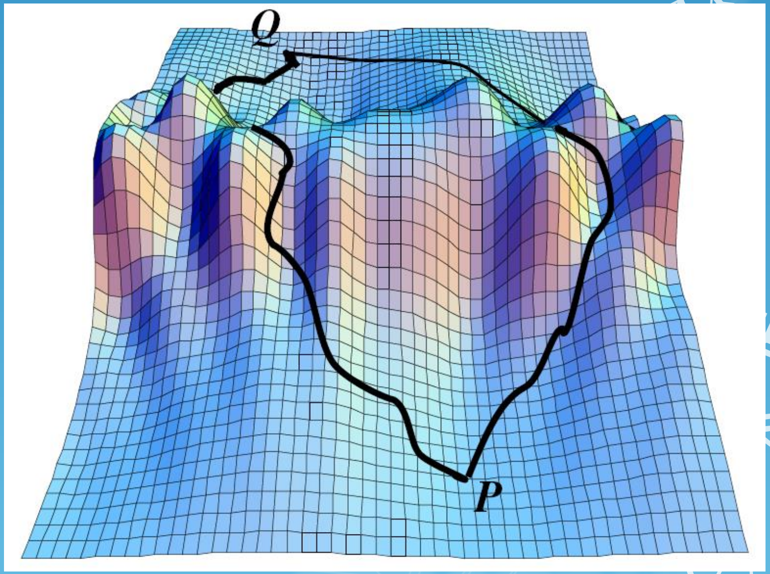
- Get time series of real data:
  - Data needs to be cleaned
- Divide the sample set in two:
  - In Sample:
    - Find optimal parameters
  - Out of Sample:
    - Test statistical robustness of the optimal parameters found In Sample

Backtesting

- Stage 4
  - Find the optimal parameters for the utility function

- What to do:
  - Find the best parameters In Sample:
    - By "brute force" if ≤ 2 parameters:
      - Plot the whole utility function
      - Find the global maximum
    - By numerical methods:
      - Gradient methods
      - Beware of
        - Saddle points
        - Local maxima

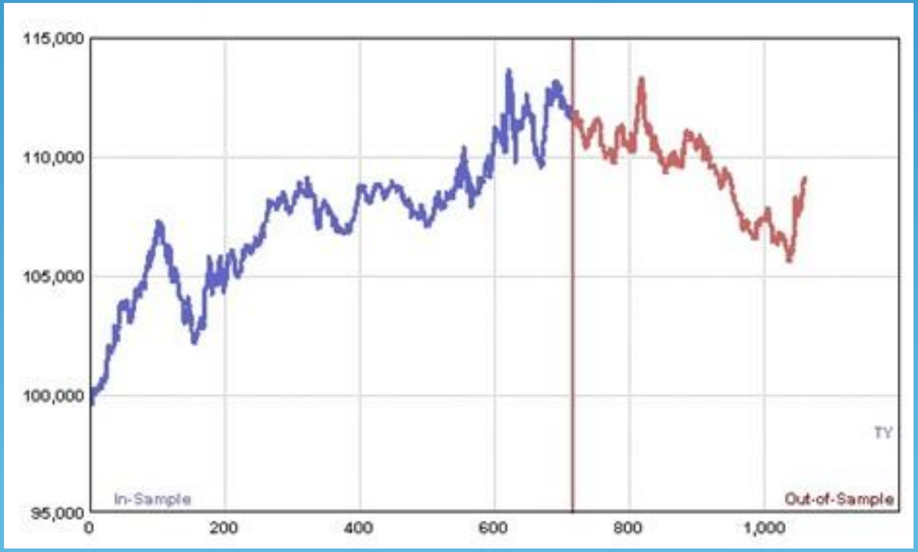
graphics.ethz.ch/~achapiro



Backtesting

- Stage 5
  - Test the stability of the In-Sample best parameters in the Out of Sample

- What to do:
  - If the pattern is still profitable, keep the strategy
  - Otherwise, discard the strategy and restart from Stage 1

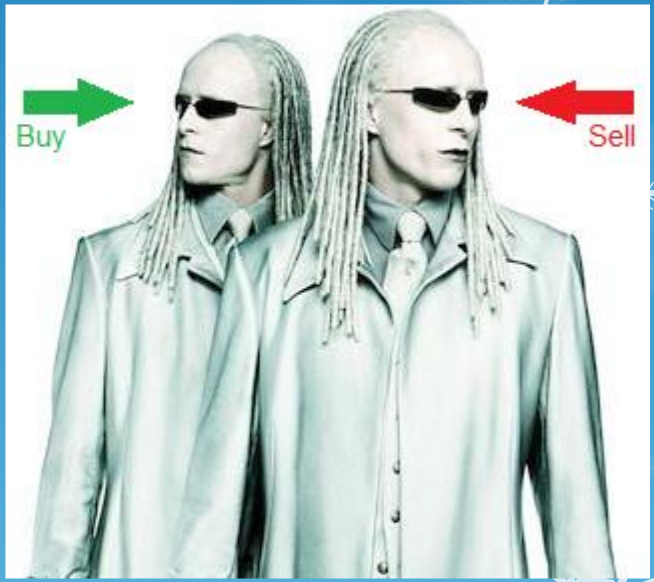


http://aostrading.cz/

3. Example: Pairs Trading

Pairs Trading

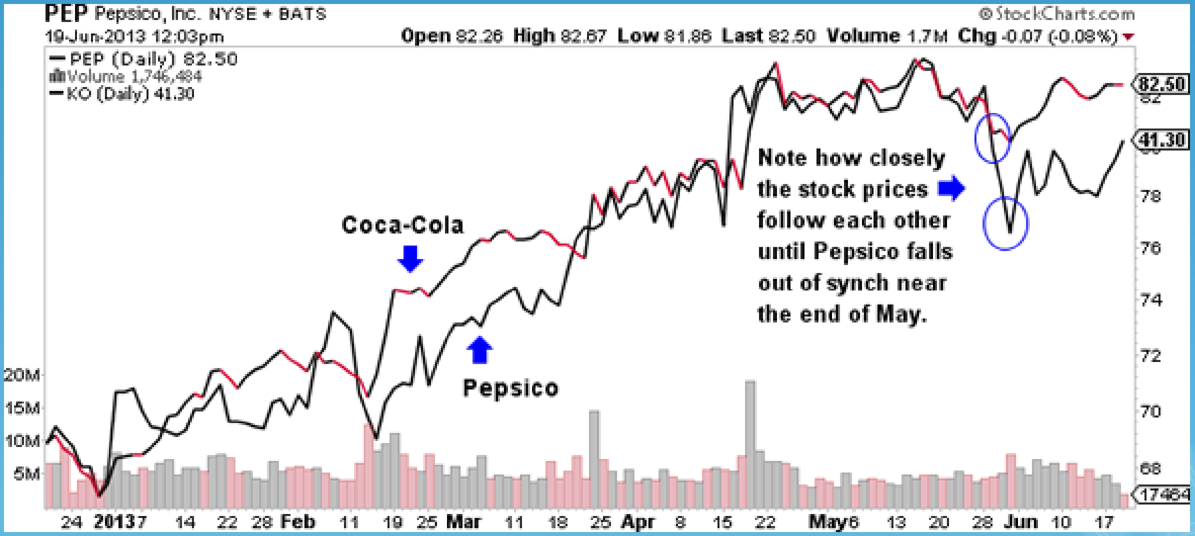
- Market neutral strategy
  - Buy one stock, sell another
  - Their correlation needs to be strong:
    - Same sector, country, market cap, etc
- Assumptions:
  - Correlation will continue to be strong
  - Any break in correlation is temporary
- The spread is mean reverting
  - Sell spread if it is large:
    - Sell outperforming stock
    - Buy underperforming stock
  - Buy spread if it is small



Film Matrix Reloaded

Pairs Trading

- Pairs trading tries to arbitrage temporary loss of correlation
  - It assumes both stocks will get correlated again
- Danger when the correlation pattern breaks
  - "Stocks fall out of sync"
  - As it happened with equities, rates and oil in the last year



nasdaq.com

Pairs Trading

- Too many parameters:
  - Returns:
    - Monthly? Daily?
    - Intraday? Every 5 mins, 10 mins, …, 1 hour?
  - Entry and Exit signals:
    - Based on rolling mean and volatility of spread
      - But how many points? 10, 20, 30,…?
      - $1/N$ weights? Exponential weights?
  - Transaction costs and execution:
    - Trading software?
    - Market and broker fees?
    - Market orders? Limit orders?
    - Latency?

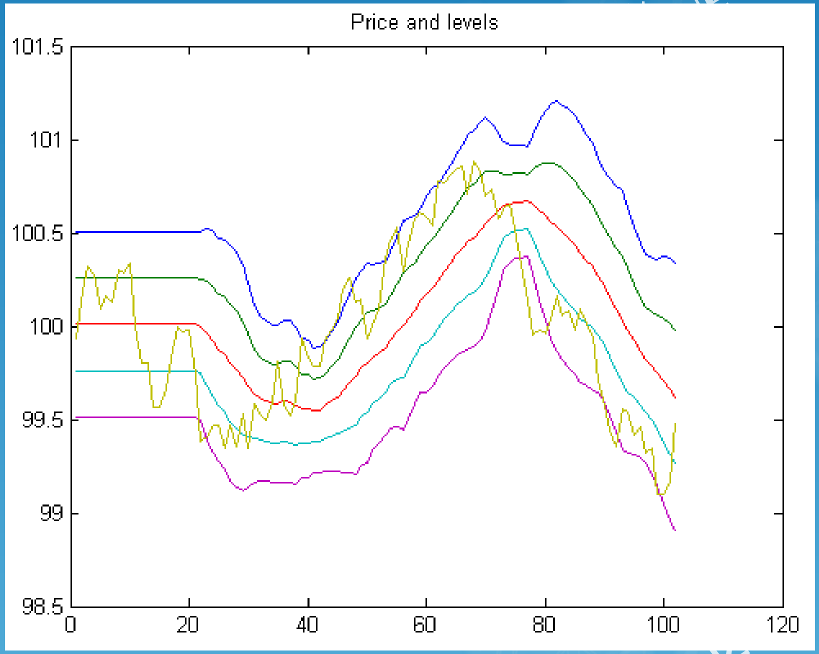
elearning.industriallogic.com

Pairs Trading

- We will use only 2 parameters:
  - The threshold of the entry signal:
    - $\mu(t) \pm a\sigma(t)$
  - The threshold of the stop-loss signal:
    - $\mu(t) \pm b\sigma(t)$, $b > a$

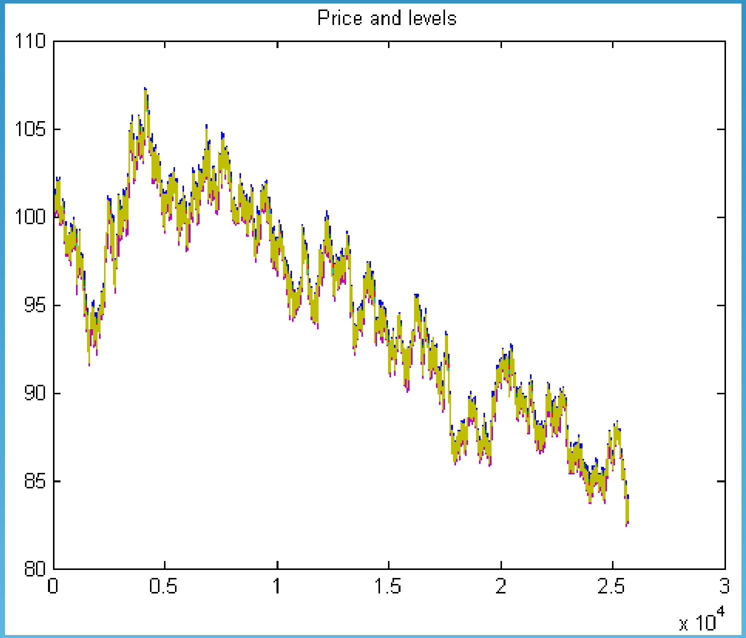
Pairs Trading: Exercise (1)

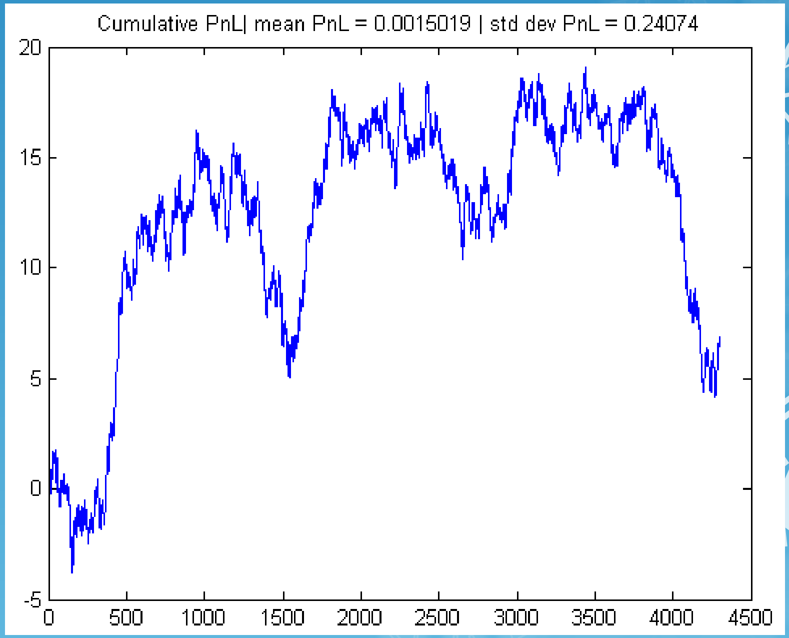
- Build the algorithm:
  - Create a price simulator
    - Use it to simulate the spread
    - Intraday prices every 5 minutes
    - One day of data
  - Compute the entry and exit levels
    - Rolling mean and volatility
  - In this exercise we used Matlab
    - You can use R, Python, etc



Pairs Trading: Exercise (2)

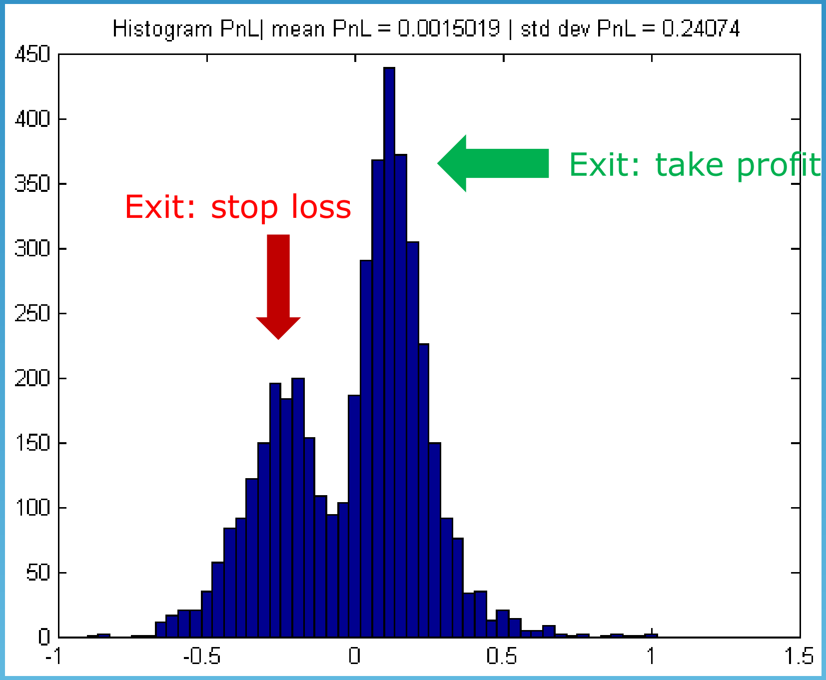
- Monte Carlo Simulation:
  - Run the algorithm for 252 days
  - Compute:
    - PNL (profit and loss) per transaction
    - Cumulative PNL





Pairs Trading: Exercise (3)

- Monte Carlo Simulation:
  - Build the distribution of PNL
    - Histogram
  - Spread was modelled as a Normal random variable
    - Shall we expect a normal distribution of PNL?



Not very "normal"!

References

- Harris, "*Trading and Exchanges*", Oxford University Press, 2003
- Barry Johnson, "*Algorithmic Trading & DMA*", 4Myeloma Press, 2010
- There are many books in Quantitative Trading
- Try some free MOOCs on <u>Machine Learning</u>:
  - Coursera (Stanford University)
  - Udacity (Georgia Tech)

LOTR Movie

Thank you for your attention